Assignment - 7

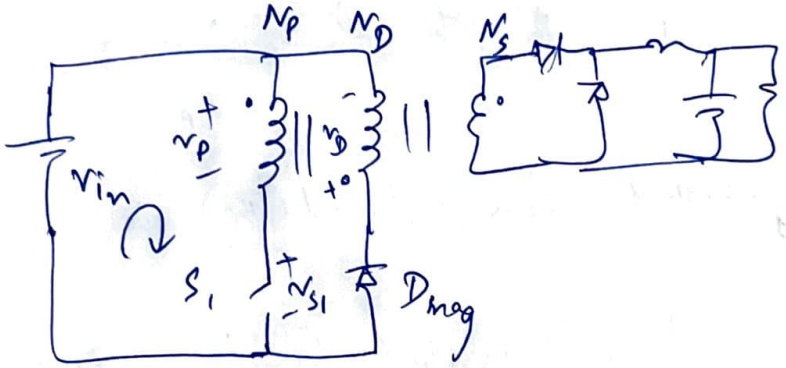
Q-1.

$$V_o = n V_{in} D \qquad \left(n = \frac{N_s}{N_p}\right)$$

$$= \frac{5}{2} \times 0.4 \times 50$$

$$V_o = 50V$$

Q-3.



When S_1 is off, the D_{mag} is ON

$$\Rightarrow V_D = -V_{in}$$

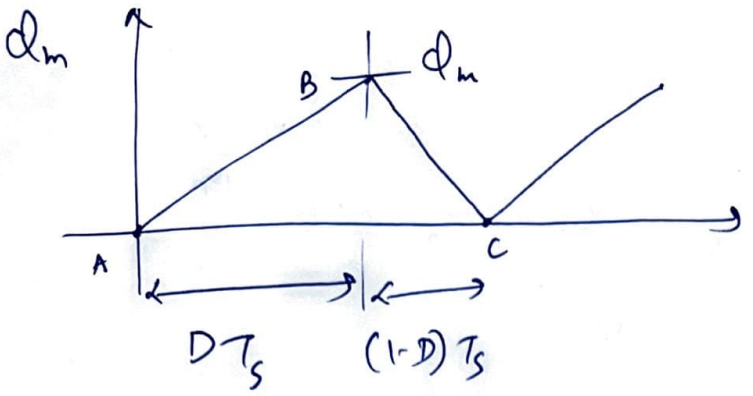
$$\Rightarrow \frac{V_p}{V_D} = \frac{N_p}{N_D}$$

$$\Rightarrow V_p = \frac{-V_{in} \times 1}{3}$$

→ apply KVL in the loop

$$\Rightarrow \boxed{V_{S1} = V_{in} + \frac{V_{in}}{3} = \frac{2}{3} V_{in}}$$

Q.4.



During flux buildup (AB), the $\phi_m = \frac{V_{in}}{N_p} \cdot DT_s$

During flux ~~direction~~ going to original point (BC), the $\phi_m = -\frac{V_{in}}{N_D}(1-D)T_s$

Now, these two values (magnitude term) should be ~~same~~ same

$$\Rightarrow \frac{V_{in}}{N_p} DT_s = \frac{V_{in}}{N_D}(1-D)T_s \qquad (\text{Given } N_D = 3N_p)$$

$$\Rightarrow \frac{D}{N_p} = \frac{1}{3N_p}(1-D)$$

$$3D = 1-D$$

$$\Rightarrow 4D = 1$$

$$\Rightarrow D = 0.25$$

$\Rightarrow$ turn on time of 'S_1', $DT_s = 0.25 \times \frac{1}{10000}$ $\left(\text{as } T_s = \frac{1}{10\,kHz}\right)$

$$= 25\,\mu s$$

$$\Rightarrow \boxed{(t_{on,S1})_{max} = 25\,\mu s}$$

Q.7.

$$E = \frac{1}{6} \times 60 \times 200 \times 200 \times 10^{-9}$$ (as both current & voltage is changing)

$$= 400 \times 10^{-6}$$

$$\boxed{E = 400\,\mu J}$$

Q.9.

→ ~~$D = \frac{20\,\mu s}{20000}$~~ $D = 20\,\mu s \times 20000$

$= 0.4$

$$V_0 = \frac{N_s}{N_p} \cdot D \cdot V_{in}$$

$$= \frac{3}{2} \times 0.4 \times 100$$

$$\boxed{V_0 = 60V}$$

Q.10.

$$L = \frac{n V_{in} D (1-D)}{2 f_{sw} \cdot \Delta i_L} \quad ; \quad I_0 = \frac{500}{100} = 5A$$

→ $$L = \frac{V_0 (1-D)}{2 f_{sw} \Delta i_L} \quad ; \quad V_0 = n V_{in} D$$

→ ~~$L = 100$~~ $100 = \frac{2}{4} \times 400 \times D$

$\Rightarrow D = 0.5$

$\Rightarrow L = \frac{V_o(1-D)}{2 f_{sw} \Delta i_L} = \frac{100 \times (1-0.5)}{2 \times 20000 \times 0.05 \times 5}$ $\quad (\Delta i_L = 5\% \text{ of } I_o)$

$\Rightarrow L = 5 \times 10^{-3}$

$\boxed{L = 5\ mH}$

Q.11. It is the soft switching case hence dv/dt depends on the rate at which Coss of switches (S_1 & S_2) changes its state

$\Rightarrow I_o = 2 C_{oss} \frac{dv}{dt}$ $\quad \left(I_o = \frac{100V}{10\Omega} = 10A\right)$

$\Rightarrow \frac{dv}{dt} = \frac{10}{2 \times 1 \times 10^{-9}}$

$\Rightarrow \boxed{\frac{dv}{dt} = 5 \ V/ns}$